Pierpaolo **Vendittelli**

Postdoctoral Researcher · AI for Digital Pathology

(+39) 371 198 8078 | pierpaolo.venditelli93@gmail.com | pierpaolov.github.io | PierpaoloV | pierpaolo-vendittelli

EXPERIENCE

Postdoc Researcher *Dec. 2024 – Present*

Prinses Maxima Centre for Pediatric Oncology *Utrecht, The Netherlands*

- Developing deep-learning and MIL pipelines for WSI analysis of Wilms tumor, targeting anaplasia grading and clinical outcome prediction.
- Coordinating multi-center dataset integration across collaborating hospitals; enforcing reproducible, clinically validated computational workflows.
- Translating model outputs into interpretable biomarkers in close collaboration with pathologists and pediatric oncologists.
- Supervising students and contributing to peer-reviewed publications in computational pathology and pediatric oncology.

PhD Candidate – AI for Digital Pathology *Mar. 2021 – Jun. 2026 (exp.)*

Radboud University Medical Center *Nijmegen, The Netherlands*

- Developed deep-learning pipelines for pancreatic cancer WSI analysis within PANCAIM (EU Horizon 2020): multi-class U-Net for TSR quantification (368 patients / 4 datasets / 25 NL labs; Dice 0.75–0.86; TSR AUC 0.61 on TCGA-PAAD). Published in *PLOS ONE* (2024) and *SPIE MI* (2022).
- Extended work to multi-tissue segmentation for combining AI-based biomarkers (TSR, mitosis density, stromal cell density, TTR) in multivariate survival analysis across four independent PDAC cohorts. Published at *COMPAYL* (MICCAI 2025 workshop).
- Co-authored multi-center clinical studies: *European Radiology* (2025) on end-to-end prognostication; *The Lancet Oncology* (2025) PANORAMA study on AI-assisted pancreatic cancer detection; *Medical Image Analysis* (2025) scoping review on multimodal AI.
- Teaching Assistant for *Image Analysis* and *Medical Imaging Analysis* across 4 course editions at Radboud University; co-supervised Master's theses.

Machine Learning Engineer *Oct. 2020 – Feb. 2021*

Everis Italy (NTT DATA) *Rome, Italy*

- Built AI chatbots and conversational systems for clients in banking and automotive.
- Annotated industrial data for object recognition systems and developed live translation pipelines for video content.

Research Intern *Feb. 2020 – Sep. 2020*

ViCOROB – NeuroImage Computing Group, University of Girona *Girona, Spain*

- Benchmarked MRI synthesis architectures (ResU-Net, cGAN, CycleGAN) on WMH and BraTS 2018 datasets for T1↔FLAIR synthesis.
- Demonstrated ResU-Net achieves superior SSIM and lower MSE vs. GAN variants under data-scarce conditions; informed Master's thesis (grade 8.6/10).

PROJECTS & PUBLICATIONS

Weekly AI Paper-Triage Agent *Personal Project*

github.com/PierpaoloV *2024 – Present*

Multi-agent pipeline (OpenAI API, Python): weekly arXiv / PubMed scan → LLM summarisation → interest-based scoring (0–10) → automated email digest. In active personal use since 2024.

Automatic quantification of tumor-stroma ratio as a prognostic biomarker for pancreatic cancer *PLOS ONE*

doi:10.1371/journal.pone.0301969 *May 2024*

Multi-class U-Net for TSR segmentation across 368 patients / 4 datasets (25 NL labs). Dice 0.75 (epithelium) and 0.86 (tumor bulk) on external TCGA-PAAD validation. TSR AUC 0.61 for 6-month survival prediction on independent cohort.

Automatic quantification of TSR as a prognostic marker for pancreatic cancer *MIDL 2023*

openreview.net/pdf?id=Dtz_iaUpGc *Sep. 2023*

Poster presentation: segmentation of tumor epithelium, healthy epithelium and TSR estimation from pancreatic WSIs.

Setting the research agenda for clinical AI in pancreatic adenocarcinoma imaging *Cancers*

doi:10.3390/cancers14143498 *Jul. 2022*

Systematic review on the current role of AI for pancreatic cancer diagnosis and personalised treatment.

Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas *SPIE MI 2022*

doi:10.1117/12.2611542 *Apr. 2022*

Multi-task U-Net with auxiliary classification head for coarse pancreatic tumour area segmentation in WSIs; classification head reduced false positives.

End-to-end prognostication in pancreatic cancer by multimodal deep learning: a retrospective, multicenter study *European Radiology*

2025

Schuurmans et al. Multimodal deep learning combining CT imaging and clinical data for PDAC outcome prediction across multiple centers.

Artificial intelligence and radiologists in pancreatic cancer detection (PANORAMA): an international observational study *The Lancet Oncology*

2025

Alves et al. International paired non-inferiority study comparing AI and radiologists for pancreatic cancer detection on standard-of-care CT.

Navigating the landscape of multimodal AI in medicine: a scoping review *Medical Image Analysis*

2025

Schouten et al. Scoping review on technical challenges and clinical applications of multimodal AI in medicine.

Combining AI-based Biomarkers for PDAC Survival *COMPAYL (MICCAI 2025 Workshop)*

Sep. 2025

Hensens, Sabroso-Lasa, Verbeke, Malats, Litjens, **Vendittelli** et al. Multi-tissue segmentation pipeline combining TSR, mitosis density, stromal cell density and TTR for multivariate PDAC survival prediction.

EDUCATION

Joint M.Sc. in Medical Imaging and Applications — *Sep. 2018 - Sep. 2020*

University of Burgundy, University of Cassino and Southern Lazio, University of Girona — *France, Italy, Spain*

- Final Grade: Bien (France), 102/110 (Italy), 8.6/10 (Spain)
- Scholarships: 2018-2020 Consortium Grant for the Erasmus Mundus Joint Master Degree.
- Relevant Courses: Computer-Aided Diagnosis, Medical Image Segmentation and Registration, Advanced Image Analysis.

B.Sc. in Informatics and Telecommunication Engineering — *Sep. 2012 - Sep. 2017*

University of Cassino and Southern Lazio — *Cassino, Italy*

- Final Grade: 94/110
- Scholarships: 2016 Erasmus+ Funded Exchange at Fontys University of Applied Sciences in Eindhoven, The Netherlands.
- Relevant Courses: Software Engineering, Databases, Mathematical Analysis, Object-Oriented Programming, Statistics.

Honors & Grants

2021 – 2024	**PhD Fellowship**, PANCAIM – Horizon 2020 EU-funded consortium for AI in pancreatic cancer	*Nijmegen, The Netherlands*
2018 – 2020	**Consortium Grant**, Erasmus Mundus Joint Master Degree in Medical Imaging and Applications (MAIA)	*France, Italy, Spain*
2016	**Erasmus+ Exchange Grant**, Funded exchange semester at Fontys University of Applied Sciences	*Eindhoven, The Netherlands*

Conference Presentations

Medical Imaging with Deep Learning (MIDL 2023) — *Nashville, TN, USA*

Poster Presentation — *Sep. 2023*

- "Automatic quantification of TSR as a prognostic marker for pancreatic cancer" – segmentation of tumor epithelium, healthy epithelium and tumor-stroma ratio from H&E WSIs.

SPIE Medical Imaging 2022 — *San Diego, CA, USA*

Poster Presentation — *Apr. 2022*

- "Automatic tumour segmentation in H&E-stained whole-slide images of the pancreas" – multi-task U-Net with classification head for pancreatic tumour area segmentation.

COMPAYL Workshop, MICCAI 2025 — *Daejeon, South Korea*

Poster Presentation — *Sep. 2025*

- "Combining AI-based Biomarkers for PDAC Survival" – multi-tissue segmentation pipeline integrating TSR, mitosis density, stromal cell density and tumor-to-tissue ratio for multivariate survival prediction in pancreatic cancer.

ECDP 2026 – European Congress of Digital Pathology — *Graz, Austria*

Poster Presentation — *Jun. 2026*

- "Automatic classification of Anaplasia in Wilms Tumor" – deep learning pipeline for automated anaplasia grading in pediatric kidney tumor WSIs.

SKILLS

Languages: Python, C++, C, Bash, SQL
ML / Deep Learning: PyTorch, scikit-learn, HuggingFace, Transformers, MONAI, Keras

MLOps & Infrastructure: Docker, Git, SLURM / GPU clusters, OpenSlide, W&B
LLM & Agents: OpenAI API, Anthropic API, prompt engineering, multi-agent orchestration

CERTIFICATIONS

- **Machine Learning**, Stanford University
- **Convolutional Neural Networks**, deeplearning.ai
- **Generative Adversarial Networks (GAN) Specialization**, deeplearning.ai
- **Improving Deep Neural Networks: Hyperparameter Tuning, Regularization & Optimization**, deeplearning.ai
- **Neural Networks and Deep Learning**, deeplearning.ai
- **AI for Medical Diagnosis**, deeplearning.ai
- **AI for Medical Prognosis**, deeplearning.ai
- **Applied Plotting, Charting & Data Representation in Python**, University of Michigan